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Working title: motion of two rigid bodies

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(Received xx; revised xx; accepted xx)

The dynamics of freely moving rigid bodies in fluid remain a fundamental challenge in fluid mechanics, with implications for natural and engineered systems. While potential flow theory predicts integrable motion for single and multiple inviscid spheres, less symmetric bodies can exhibit far richer behaviour. In this work, we investigate the onset of chaotic dynamics in pairs of immersed ellipsoids, combining potential flow simulations based on boundary element methods with selected direct numerical simulations (DNS) of viscous flow. Systematic variation of density ratio, energy ratio, and initial separation reveals that spheres exhibit strictly periodic motion, whereas spheroids and triaxial ellipsoids develop quasi-periodic and chaotic trajectories due to rotational–translational coupling and geometric asymmetry. Departures from sphericity strongly accelerate the transition to chaos, with heterogeneous pairs (sphere + ellipsoid) triggering irregular motion at lower thresholds than homogeneous configurations. Complementary DNS at moderate Reynolds numbers shows that viscosity damps short-term motion but enhances long-term irregularity through vortex shedding. These results provide new insight into how body geometry and flow conditions govern the emergence of chaos in inviscid and viscous regimes, with relevance to particle-laden flows, sediment dynamics, and bio-inspired propulsion.

1. Introduction

The interaction between rigid bodies and surrounding fluid is a central theme in fluid mechanics, relevant to problems ranging from sediment transport in geophysics to locomotion in biology and the design of bio-inspired devices. While much progress has been made in modelling and predicting these dynamics, the motion of freely moving bodies in inviscid flow remains an active research area. In particular, the emergence of chaotic behaviour in such nominally simple regimes challenges conventional assumptions and highlights the subtle interplay between geometry, inertia, and flow topology.

Ellipsoidal bodies provide a versatile framework for investigating these phenomena. Their geometry is sufficiently streamlined to permit analytical treatment, yet sufficiently general to exhibit rich dynamical responses. Classical studies of ellipsoids in inviscid flow, dating back

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to Kirchhoff’s formulation of rigid body motion in potential flow Kirchhoff (1876) and Lamb’s closed-form derivation of hydrodynamic forces for ellipsoids Lamb (1945), revealed that the problem reduces to a set of coupled ordinary differential equations. These formulations have underpinned subsequent developments in the theory of added mass and potential flow. More recent work has demonstrated that even in the absence of viscosity and vorticity diffusion, ellipsoids can display complex trajectories, including chaos, when asymmetries and multi-body interactions are introduced.

The present study investigates the chaotic dynamics of freely moving ellipsoids in inviscid flow, combining potential flow theory with numerical simulations and dynamical systems analysis. By systematically varying density ratio, energy ratio, and body geometry, we identify the regimes in which trajectories transition from regular to chaotic. Particular emphasis is placed on the role of geometric asymmetry: while spheres yield integrable dynamics, spheroids and triaxial ellipsoids introduce sufficient complexity to generate irregular and unpredictable motion. Complementary simulations using direct numerical simulation (DNS) in viscous flow provide additional context for assessing the influence of viscosity.

2. Direct Numerical Simulations (DNS) using the X3D-NDOF solver

We employ two complementary approaches to model the dynamics of immersed rigid bodies. First, in this section, we develop and validate a direct numerical simulation (DNS) capability for fully resolved immersed rigid bodies with six degrees of freedom (6DOF), built on the high-order, parallel Xcompact3D framework (X3D-NDOF) [The following citation needs fixing] et al (2005). We solve the full Navier–Stokes equations by an immersed boundary method, along with solving for the dynamics of the moving bodies. The objective is to capture complex fluid–structure interactions with moving boundaries without mesh deformation or re-meshing. Below, we outline the governing equations, the non-dimensionalisation, and the sixth-order compact finite-difference discretisation; describe the time-integration strategy and an immersed boundary method (IBM) enforcing no-slip conditions on arbitrarily moving bodies; and present the force/torque evaluation and solid-motion coupling.

Second, in Section 3, a potential-flow solver based on a boundary element method (BEM) is used to obtain inviscid interactions between the bodies with high accuracy and low computational cost. Together, these methods provide a consistent basis for differentiating inviscid and viscous effects, with cross-validation ensuring robustness of the results. Both solvers are validated against canonical test cases, including flow past a sphere, settling under gravity, and Jeffery’s orbits in shear flow.

2.1. Fluid solver: equations and methodology

The fluid is assumed incompressible and Newtonian; the immersed solid is rigid and non-porous. The governing equations are

$$\nabla \cdot \mathbf{u} = 0, \quad (2.1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \left[\nabla(\mathbf{u} \otimes \mathbf{u}) + \frac{1}{2}(\mathbf{u} \cdot \nabla)\mathbf{u} \right] = -\frac{1}{\rho_f} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}. \quad (2.2)$$

Here ν is the kinematic viscosity (setting $\nu = 0$ results in Euler’s equations for inviscid fluid), ρ_f is the fluid density, p the pressure, and $\mathbf{f}$ an IBM forcing that enforces the boundary conditions.

2.1.1. Non-dimensionalised equations

Introducing characteristic scales, we have

$$x = L_0 x^*, \quad t = \frac{L_0}{U_0} t^*, \quad \mathbf{u} = U_0 \mathbf{u}^*, \quad p = \rho_f U_0^2 p^*,$$

where the characteristic length scale L_0 is the major axis of an immersed body, and the characteristic velocity U_0 is the relative velocity [relative to what?] of the body in the flow. The continuity equation becomes

$$\nabla^* \cdot \mathbf{u}^* = 0, \quad (2.3)$$

and the momentum equation reads

$$\frac{\partial \mathbf{u}^*}{\partial t^*} + \nabla^* \cdot (\mathbf{u}^* \otimes \mathbf{u}^*) + \frac{1}{2} (\mathbf{u}^* \cdot \nabla^*) \mathbf{u}^* = -\nabla^* p^* + \frac{1}{Re} \nabla^{*2} \mathbf{u}^* + \mathbf{f}^*, \quad (2.4)$$

with the Reynolds number

$$Re \equiv \frac{UL}{\nu}. \quad (2.5)$$

In the inviscid limit we have $1/Re = 0$.

2.1.2. Spatial discretisation

Convective and diffusive terms are discretised with sixth-order compact finite differences in Xcompact3D, achieving quasi-spectral accuracy with a moderate stencil width and compatibility with free-slip, no-slip, and periodic boundary conditions. In the simulations reported here, the outer fluid domain follows periodic boundary conditions; and at the immersed solid–fluid interface no-slip, and no penetration when $\nu \neq 0$, are enforced.

2.1.3. Time integration

We employ a three-step fractional-step scheme:

$$\frac{\mathbf{u}^* - \mathbf{u}^k}{\Delta t} = a_k \mathbf{F}^k + b_k \mathbf{F}^{k-1} - c_k \nabla \tilde{p}^k + c_k \tilde{\mathbf{f}}^{k+1}, \quad (2.6)$$

$$\frac{\mathbf{u}^{**} - \mathbf{u}^*}{\Delta t} = c_k \nabla \tilde{p}^k, \quad (2.7)$$

$$\frac{\mathbf{u}^{k+1} - \mathbf{u}^{**}}{\Delta t} = -c_k \nabla \tilde{p}^{k+1}, \quad (2.8)$$

where

$$\mathbf{F}^k = -\mathbf{u} \cdot \nabla \mathbf{u} + \nu \nabla^2 \mathbf{u}, \quad (2.9)$$

and

$$\tilde{p}^{k+1} = \frac{1}{c_k \Delta t} \int_{t_k}^{t_{k+1}} p \, dt, \quad \tilde{\mathbf{f}}^{k+1} = \frac{1}{c_k \Delta t} \int_{t_k}^{t_{k+1}} \mathbf{f} \, dt, \quad (2.10)$$

with $\mathbf{u}^*$ and $\mathbf{u}^{**}$ the predicted and intermediate velocities. The intermediate velocity from (2.7) does not satisfy (2.1), so the pressure update (2.8) enforces incompressibility for $\mathbf{u}^{k+1}$. Coefficients $a_k, b_k, c_k = a_k + b_k$ correspond to the time scheme on n_k substeps; we use third-order Adams–Bashforth.

2.2. Immersed boundary method

We adopt an alternating-direction reconstruction IBM (ADR-IBM) (Giannenas & Laizet 2021) based on 1D cubic-spline reconstructions along each coordinate direction to impose

the target wall velocity at immersed boundaries, avoiding spurious oscillations and remaining compatible with Xcompact3D's 2D domain decomposition. The forcing term is

$$c_k \tilde{\mathbf{f}}^{k+1} = \chi \left(\frac{\mathbf{u}_{re} - \mathbf{u}^k}{\Delta t} - a_k \mathbf{F}^k - b_k \mathbf{F}^{k-1} + c_k \nabla \tilde{p}^k \right), \quad (2.11)$$

where $\mathbf{u}_{re}$ is the reconstructed velocity inside the solid and

$$\chi = \begin{cases} 1, & x \in \Omega_s, \\ 0, & x \in \Omega_f, \end{cases}$$

is a solid/fluid indicator. Because $\mathbf{u}_{re}$ is not divergence-free, we solve a modified Poisson equation,

$$\nabla^2 \tilde{p}^{k+1} = \frac{\nabla \cdot [(1 - \chi) \mathbf{u}^{**} + \chi \mathbf{u}_{re}]}{c_k \Delta t}, \quad (2.12)$$

and, substituting (2.11) into (2.6), we obtain

$$\frac{\mathbf{u}^{**} - \mathbf{u}^k}{\Delta t} = \frac{\mathbf{u}_{re} - \mathbf{u}^k}{\Delta t} \iff \mathbf{u}^{**} = \mathbf{u}_{re}. \quad (2.13)$$

Equation (2.12) ensures incompressibility while recovering Laplace's equation inside Ω_s ($\chi = 1$) and the standard pressure Poisson equation in Ω_f .

For fully immersed bodies, the implementation enforces an even number of crossings [crossings of what?] in each of the x -, y -, and z -directions. The velocity field in the solid is reconstructed via cubic splines to meet the wall velocity at immersed interfaces while maintaining continuity across the domain. Further details of this can be seen in Giannenas & Laizet (2021).

2.3. Solid motion solver in Xcompact3D and coupling

2.3.1. Force evaluation

For a body with surface S_H , the hydrodynamic force is

$$\mathbf{F}_A = \int_{S_H} \mathbf{T} dS, \quad (2.14)$$

where $\mathbf{T} = \boldsymbol{\tau} \cdot \mathbf{n}$ and

$$\tau_{ij} = -p \delta_{ij} + 2\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right). \quad (2.15)$$

Drag and lift coefficients (when relevant) follow as

$$C_D = \frac{4}{\rho U_\infty^2 \pi R^2} \mathbf{F}_A \cdot \mathbf{e}_X, \quad (2.16)$$

$$C_L = \frac{4}{\rho U_\infty^2 \pi R^2} \mathbf{F}_A \cdot \mathbf{e}_Y. \quad (2.17)$$

A control volume V bounded by $S = S_I + S_O + S_C + S_F + S_L + S_R + S_B$ yields, from momentum conservation,

$$\frac{D}{Dt} \int_V \rho \mathbf{u} dV = \frac{\partial}{\partial t} \int_V \rho \mathbf{u} dV + \int_V \rho \mathbf{u} \cdot \nabla \mathbf{u} dV = \int_S \mathbf{T} dS, \quad (2.18)$$

144 and, via the transport theorem,

$$145 \quad \frac{\partial}{\partial t} \int_V \rho \mathbf{u} dV + \underbrace{\int_S \rho \mathbf{u} (\mathbf{u} \cdot \mathbf{n}) dS}_A = \underbrace{\int_S \mathbf{T} dS}_B. \quad (2.19)$$

146 Surface integrals over the cuboidal faces are evaluated accurately; the contribution of A over
 147 the body surface vanishes due to non-penetration ($\mathbf{u} \cdot \mathbf{n} = 0$). Rearrangement isolates the
 148 desired integral of $\mathbf{T}$ over the body surface.

149 2.3.2. Torque evaluation

150 The total torque is

$$151 \quad \boldsymbol{\tau} = \int_{S_B} \mathbf{r} \times \mathbf{T} dS. \quad (2.20)$$

152 Angular momentum conservation gives

$$153 \quad \frac{D}{Dt} \int_V \mathbf{r} \times (\rho \mathbf{u}) dV = \frac{\partial}{\partial t} \int_V \mathbf{r} \times (\rho \mathbf{u}) dV + \int_V \mathbf{r} \times (\rho \mathbf{u} \cdot \nabla \mathbf{u}) dV = \int_S \mathbf{r} \times \mathbf{T} dS, \quad (2.21)$$

154 or, using the transport theorem,

$$155 \quad \frac{\partial}{\partial t} \int_V \mathbf{r} \times (\rho \mathbf{u}) dV + \int_S (\mathbf{r} \times \rho \mathbf{u}) (\mathbf{u} \cdot \mathbf{n}) dS = \int_S \mathbf{r} \times \mathbf{T} dS. \quad (2.22)$$

156 All calculations are performed in the body's frame of reference. The rigid-body dynamics
 157 are advanced as outlined below.

158 2.4. Predictor-Corrector Direct Multiplier Method

159 The algorithm used to step forward the orientation of the ellipsoid is presented here. We will
 160 represent the orientation of the body at time n as q_n , and the angular velocity of the body at
 161 time n as ω_n .

162 Firstly, the initial angular velocity and torque are transformed into the body frame from
 163 the lab frame as follows:

$$164 \quad \omega_n^b = q_n^* \omega_n q_n \quad (2.23)$$

$$166 \quad \tau_n^b = q_n^* \tau_n q_n. \quad (2.24)$$

167 The angular velocity is then calculated at the half- and quarter-time step in the body frame:

$$168 \quad \omega_{n+\frac{1}{4}}^b = \omega_n^b + \frac{1}{4} \dot{\omega}_n^b \delta t \quad (2.25)$$

$$170 \quad \omega_{n+\frac{1}{2}}^b = \omega_n^b + \frac{1}{2} \dot{\omega}_n^b \delta t, \quad (2.26)$$

171 where the angular acceleration $\dot{\omega}_n^b$ is calculated from the torque. The quarter-time step angular
 172 velocity is then converted back to lab frame and used to make a *prediction* of the orientation
 173 quaternion q at the half-time interval:

$$174 \quad \omega_{n+\frac{1}{4}} = q_n \omega_{n+\frac{1}{4}}^b q_n^*, \quad (2.27)$$

$$176 \quad q'_{n+\frac{1}{2}} = \left[\cos \left(\frac{\|\omega_{n+\frac{1}{4}}\| \delta t}{4} \right), \sin \left(\frac{\|\omega_{n+\frac{1}{4}}\| \delta t}{4} \right) \frac{\omega_{n+\frac{1}{4}}}{\|\omega_{n+\frac{1}{4}}\|} \right] q_n. \quad (2.28)$$

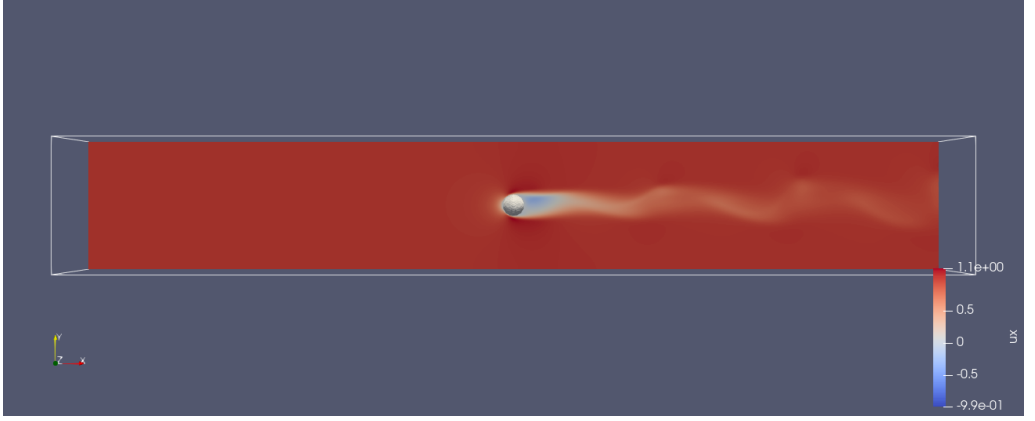


Figure 1: Flow velocity magnitude around a stationary sphere, at a Reynolds number of 300. The computed drag compares very well with previous literature, see Table 1.

177 This prediction $q'_{n+\frac{1}{2}}$ is used to calculate the angular velocity $\omega_{n+\frac{1}{2}}$ as follows:

$$178 \quad \omega_{n+\frac{1}{2}} = q'_{n+\frac{1}{2}} \omega_{n+\frac{1}{2}}^b q'_{n+\frac{1}{2}}^* \quad (2.29)$$

179 Thus, the correct unit quaternion q_{n+1} is as follows:

$$180 \quad q_{n+1} = \left[\cos\left(\frac{\|\omega_{n+\frac{1}{2}}\| \delta t}{2}\right), \sin\left(\frac{\|\omega_{n+\frac{1}{2}}\| \delta t}{2}\right) \frac{\omega_{n+\frac{1}{2}}}{\|\omega_{n+\frac{1}{2}}\|} \right] q_n. \quad (2.30)$$

181 Finally, the angular velocity in body frame can be calculated and transformed to lab frame:

$$182 \quad \omega_{n+1}^b = \omega_n^b + \dot{\omega}_{n+\frac{1}{2}}^b \delta t, \quad (2.31)$$

183

$$184 \quad \omega_{n+1} = q_{n+1} \omega_{n+1}^b q_{n+1}^*. \quad (2.32)$$

185 2.5. Validation against simple solid motion at finite Reynolds number

186 2.5.1. Hydrodynamic drag on a fixed and moving sphere

187 We simulate flow past a fixed sphere at $Re_p = 300$ in a domain $L \times H \times W = 40D \times 6D \times 6D$
 188 with $n_x \times n_y \times n_z = 1025 \times 256 \times 256$. A uniform inflow/outflow of $U_0 = 1.0$ is imposed,
 189 and periodic conditions apply in y and z . The initial field is perturbed with Gaussian noise
 190 in y, z (peak 0.1% of U_0). The sphere is centred; the run spans $t = 250$ (non-dimensional)
 191 with $\Delta t = 7.5 \times 10^{-4} D/u_\infty$. A snapshot is shown in Figure 1.

192 We report the mean drag coefficient $C_D = 2F_D/(\rho U^2 A)$ and Strouhal number $St = fD/U$.
 193 The results ($\overline{C_D} = 0.640$, $St = 0.136$) in Table 1 agree closely with previous studies R &
 194 S (1997); Wieselberger (1922); et al (2008); Henderson (1995); Williamson (1992, 1996);
 195 Clair *et al.* (1970); Roos & Willmarth (1971); Johnson & Patel (1999); Ploumhans *et al.*
 196 (2002). Repeating the case with $u_\infty = 0.5$ and sphere velocity $u_x = -0.5$ (same relative
 197 velocity) preserves accuracy for moving boundaries.

198 2.5.2. Validation against a 6DOF settling sphere at $Re_p = 300$

199 A simulation of a sphere settling in a viscous fluid was conducted to assess the accuracy of the
 200 linear force evaluation and solid motion solver. This canonical problem has been extensively

Study	$\overline{C_D}$	St
Clair <i>et al.</i> (1970)	0.632	–
Roos & Willmarth (1971)	0.629	–
Johnson & Patel (1999)	0.656	0.137
Kim <i>et al.</i> (2001)	0.657	0.134
Ploumhans <i>et al.</i> (2002)	0.683	0.135
Constantinescu & Squires (2003)	0.655	0.136
Constantinescu & Squires (2004)	0.655	–
Campregher <i>et al.</i> (2009)	0.675	–
Mimeau <i>et al.</i> (2016)	0.673	0.133
X3D-NDOF Fixed	0.644	0.136
X3D-NDOF Moving	0.624	0.136

Table 1: Comparison of mean drag coefficient $\overline{C_D}$ and Strouhal number St at $Re_p = 300$.

studied and provides a reliable benchmark for validation. We define the effective gravity

$$\mathbf{g}_{\text{eff}} = \frac{\rho_s - \rho_f}{\rho_s} \mathbf{g}. \quad (2.33)$$

As a particle accelerates, F_d grows until it balances $\mathbf{g}_{\text{eff}}$, yielding terminal velocity U_t and particle Reynolds number

$$Re_p = \frac{\rho_f U_t D}{\mu}. \quad (2.34)$$

For $Re_p \ll 1$, $F_d = 6\pi\mu U_t R$ and

$$U_{t0} = \frac{2(\rho_s - \rho_f)}{\mu} g R^2. \quad (2.35)$$

At low–moderate Re_p , a more suitable drag correlation is that of Clift *et al.* (2005)

$$C_d = C_{d0} + \frac{4}{Re_p^{0.33}}, \quad (2.36)$$

leading to

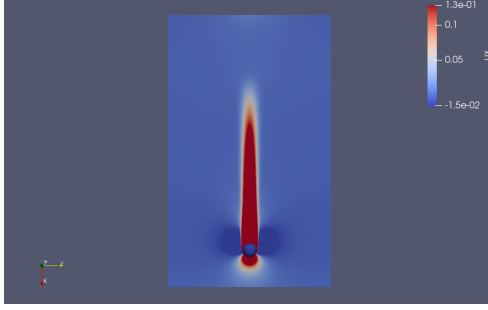
$$U_t = \sqrt{\frac{d(\rho_s - \rho_f)gRe_p}{3\rho_f(6 + Re_p^{0.67})}}. \quad (2.37)$$

We use a domain $12D \times 40D \times 12D$ with $512 \times 1024 \times 512$ cells and periodic boundaries. A sphere of diameter D and $\rho_s = 2\rho_f$ is placed at $(6D, 35D, 6D)$ and settles under gravity (Figure 2a). The velocity history (Figure 2b) agrees well with theory; Appendix ?? reports additional Re_p cases with similarly good agreement.

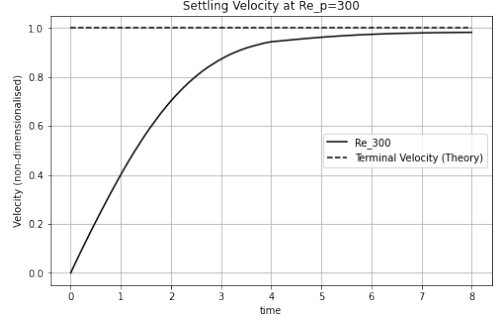
2.5.3. Rotational motion: Jeffery’s orbits

In figure 3, we consider a neutrally buoyant ellipsoid in Couette flow with shear rate γ ; top/bottom walls move, other directions are periodic For a rotationally symmetric ellipsoid with axes (a, b, b) (i.e., a spheroid), we define

$$\epsilon = \frac{a}{b}. \quad (2.38)$$

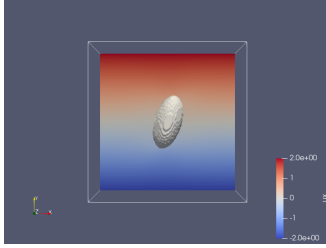


(a) Sphere settling under gravity at $Re_p = 300$.

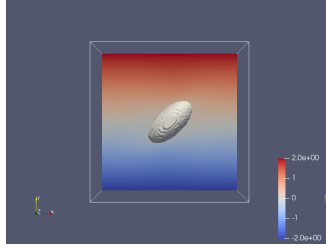


(b) Settling velocity of a sphere at $Re_p = 300$.

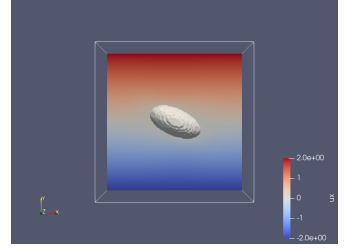
Figure 2: Simulation and validation of a sphere settling in a viscous fluid at $Re_p = 300$.



(a) $t = 0.2$



(b) $t = 0.8$



(c) $t = 1.5$

Figure 3: Shear-flow velocity around a rotating ellipsoid at different times.

221 The theory of Jeffery (1922) gives the orientation dynamics of a sphere placed in a constant
 222 shear at low Re_p . Taking a as the major-axis length and $U_0 = \gamma a$, the particle Reynolds
 223 number is

$$224 \quad Re_p = \frac{\rho_f \gamma a^2}{\mu}. \quad (2.39)$$

225 For a sphere, the rotation rate is

$$226 \quad \Omega = \frac{\gamma}{2}, \quad (2.40)$$

227 with period

$$228 \quad T_1 = \frac{4\pi}{\gamma}. \quad (2.41)$$

229 For general ϵ ,

$$230 \quad T_\epsilon = \frac{2\epsilon^2 + 1}{2\epsilon} T_1, \quad \theta(t) = \arctan\left(\epsilon \tan \frac{2\pi t}{T_\epsilon}\right). \quad (2.42)$$

231 We simulate a domain $4a \times 4a \times 4a$ at $Re_p = 0.2$. A good agreement with Jeffery's prediction
 232 is shown in Figure 4.

233 2.6. Computational Efficiency

234 Figure 5 shows mean time per time step grows approximately linearly with body count at fixed
 235 resolution. The dominant cost arises from per-body hydrodynamic force/torque evaluation,
 236 which scales linearly with the number of particles, confirming efficient multi-body scalability

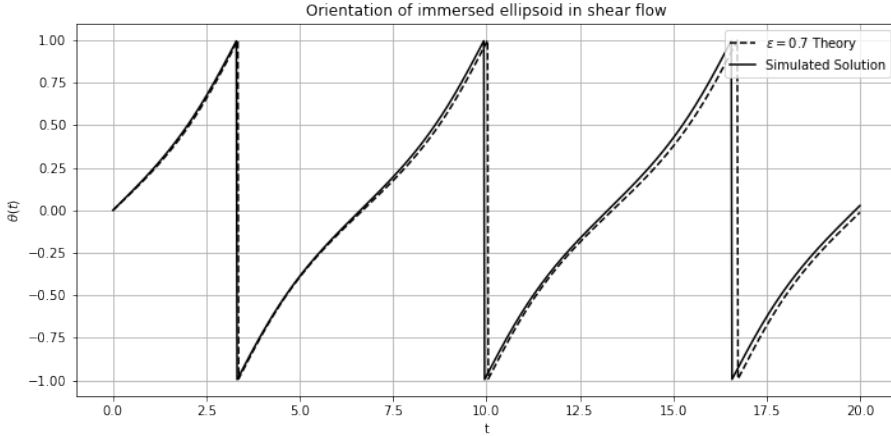


Figure 4: Orientation of a single body in shear flow over time versus Jeffery’s theory.

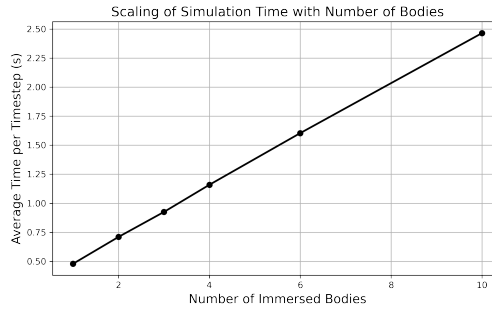


Figure 5: Computational scaling of X3D-NDOF as the number of immersed bodies increases at fixed grid resolution.

237 in X3D-NDOF under fixed spatial resolution. See (Giannenas & Laizet 2021) for the excellent
 238 performance of X3D’s current parallel efficiency.

239 3. POISE — Potential flow solver for Objects Immersed in Fluid using Surface 240 Elements

241 We develop POISE (Potential flow solver for Objects Immersed in fluid using Surface
 242 Elements), a boundary-integral framework for modelling the motion of multiple rigid
 243 ellipsoids in an inviscid fluid under potential-flow assumptions. The formulation is based on
 244 the extended Kirchhoff–Clebsch equations and is discretised via a boundary element method
 245 (BEM) on the surfaces of the immersed bodies. Six-node curved triangular surface elements
 246 are employed to represent geometry accurately and to evaluate surface integrals. Rigid-body
 247 motion in six degrees of freedom (6DOF) is advanced using a predictor–corrector scheme;
 248 hydrodynamic forces and torques are obtained from surface pressure via Bernoulli’s relation.
 249 We summarise the boundary-integral formulation, the numerical discretisation, and the time-
 250 integration strategy, before presenting verification and validation for single- and multi-body
 251 configurations.

3.1. Boundary integral equations

We use a boundary element method (BEM) to solve for the velocity potential. Rather than a volumetric discretisation, we discretise only the ellipsoid surfaces and recast the governing Laplace equation as a surface-integral problem. A standard vector-calculus identity reads

$$\phi \nabla^2 f = \nabla \cdot (\phi \nabla f) - \nabla \phi \cdot \nabla f.$$

Interchanging f and ϕ and subtracting gives, for f and ϕ satisfying Laplace's equation,

$$\nabla \cdot (\phi \nabla f - f \nabla \phi) = 0.$$

Let $f(\mathbf{x}) = G(\mathbf{x}; \mathbf{x}_0)$ be the free-space Green's function of the three-dimensional Laplace operator, with $\nabla^2 G(\mathbf{x}; \mathbf{x}_0) = \delta(\mathbf{x} - \mathbf{x}_0)$ and $G(\mathbf{x}; \mathbf{x}_0) = \frac{1}{4\pi|\mathbf{x} - \mathbf{x}_0|}$. Integrating over a domain and applying the divergence theorem yields the boundary-integral equation (BIE)

$$\begin{aligned} \phi(\mathbf{x}_0) = & - \iint G(\mathbf{x}_0; \mathbf{x}) [\mathbf{n}(\mathbf{x}) \cdot \nabla \phi(\mathbf{x})] dS(\mathbf{x}) \\ & + \iint \phi(\mathbf{x}) [\mathbf{n}(\mathbf{x}) \cdot \nabla G(\mathbf{x}_0; \mathbf{x})] dS(\mathbf{x}), \end{aligned} \quad (3.1)$$

where $\mathbf{x}_0$ lies on the boundary.

Equation (3.1) expresses the potential as a distribution of *poles* and *dipoles* over the surface, with densities $\phi(\mathbf{x})$ and $[\mathbf{n}(\mathbf{x}) \cdot \nabla G(\mathbf{x}_0; \mathbf{x})]$, respectively. The quantity $[\mathbf{n}(\mathbf{x}) \cdot \nabla G(\mathbf{x}_0; \mathbf{x})]$ is specified by boundary conditions, and the integral equation reduces to a linear system for the unknown $\phi(\mathbf{x})$. The potential at interior points follows by evaluating (3.1) off the surface.

To obtain the surface gradient, we may choose $f(\mathbf{x}) = \nabla G(\mathbf{x}; \mathbf{x}_0)$ and derive the *gradient form*

$$\begin{aligned} -\frac{\partial \phi(\mathbf{x}_0)}{\partial x_i} = & \iint_S \phi(\mathbf{x}) \nabla \left(\frac{\partial G(\mathbf{x}; \mathbf{x}_0)}{\partial x_i} \right) dS \\ & - \iint_S \frac{\partial G(\mathbf{x}; \mathbf{x}_0)}{\partial x_i} \nabla \phi(\mathbf{x}) dS. \end{aligned} \quad (3.2)$$

The kernel $\nabla (\partial G / \partial x_i)$ is $O(r^{-3})$ as $r = |\mathbf{x} - \mathbf{x}_0| \rightarrow 0$, rendering the integral *hypersingular*; evaluating it accurately requires specialised techniques. In practice, we recover surface velocity by interpolating ϕ over each boundary element and differentiating the interpolation numerically. The pressure then follows from Bernoulli's relation,

$$P + \rho \|\mathbf{u}\|^2 + \rho gh = C,$$

with gravity absent in the present setting. The hydrodynamic force and torque are

$$\mathbf{f} = \iint_S P(\mathbf{x}) \mathbf{e} \cdot \mathbf{n}(\mathbf{x}) dS(\mathbf{x}), \quad (3.3)$$

$$\mathbf{t} = \iint_S P(\mathbf{x}) \mathbf{e} \times \mathbf{n}(\mathbf{x}) dS(\mathbf{x}), \quad (3.4)$$

where $\mathbf{e} = \mathbf{x} - \mathbf{c}$ and $\mathbf{c}$ is the body centroid.

3.2. Discretisation

Following Pozrikidis (2002), we discretise the boundary as follows. Let the surface S enclosing the control volume be divided into N boundary elements E_i ($i = 1, \dots, N$).

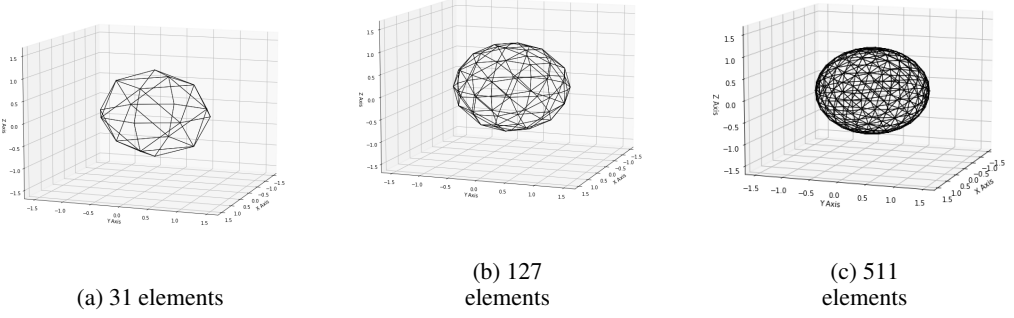


Figure 6: Discretisation of a single ellipsoid at increasing mesh resolutions.

285 Replacing integrals over S by sums over elements yields

$$\begin{aligned}
 286 \quad \phi(x_0) = & - \sum_{i=1}^N \int_{E_i} G(x, x_0) [\mathbf{n}(x) \cdot \nabla \phi(x)] dS(x) \\
 287 \quad & + \sum_{i=1}^N \int_{E_i} \phi(x) [\mathbf{n}(x) \cdot \nabla G(x, x_0)] dS(x), \quad (3.5)
 \end{aligned}$$

288 with the unit normal $\mathbf{n}$ pointing into the control region enclosed by S .

289 Each body is meshed from a tetrahedral seed whose triangular faces are recursively refined
 290 by connecting edge midpoints (four sub-triangles per refinement), mapped onto the unit
 291 sphere, and then affinely transformed to the target ellipsoid. Each boundary element is a
 292 six-node curved triangle (three vertices and three mid-edge nodes). Figure 6 illustrates the
 293 procedure.

294 3.2.1. Point collocation

295 We compute the unknown element values of ϕ or its normal derivative by enforcing (3.5) at
 296 the midpoint of each boundary element, $\mathbf{x}_j^M$ ($j = 1, \dots, N$):

$$\begin{aligned}
 297 \quad \phi(\mathbf{x}_j^M) = & -2 \sum_{i=1}^N \left(\frac{\partial \phi}{\partial n} \right)_i \int_{E_i} G(\mathbf{x}, \mathbf{x}_j^M) dS(\mathbf{x}) + 2 \sum_{i=1}^N \phi_i \int_{E_i}^{PV} [\mathbf{n}(\mathbf{x}) \cdot \nabla G(\mathbf{x}, \mathbf{x}_j^M)] dS(\mathbf{x}), \quad (3.6)
 \end{aligned}$$

298 Setting $\phi(\mathbf{x}_j^M) = \phi_j$ and writing $\phi_i = \delta_{ij} \phi_j$ leads to

$$\begin{aligned}
 299 \quad \left(A_{ij} - \frac{1}{2} \delta_{ij} \right) \phi_i = & B_{ji} \left(\frac{\partial \phi}{\partial n} \right)_i, \quad (3.7)
 \end{aligned}$$

300 with coefficient matrices

$$\begin{aligned}
 301 \quad A_{ij} \equiv & \int_{E_i} G(\mathbf{x}, \mathbf{x}_j^M) dS(\mathbf{x}), \quad (3.8)
 \end{aligned}$$

$$\begin{aligned}
 303 \quad B_{ji} \equiv & \int_{E_i} [\mathbf{n}(\mathbf{x}) \cdot \nabla G(\mathbf{x}, \mathbf{x}_j^M)] dS(\mathbf{x}) \quad (3.9)
 \end{aligned}$$

304 Given either the element values ϕ_i or the normal fluxes $(\partial \phi / \partial n)_i$, the other set follows
 305 immediately.

3.2.2. Six-node curved triangles

To capture surface curvature, each boundary element is a six-node curved triangle (Figure 7). The physical element is mapped from a reference right isosceles triangle in the $\xi\eta$ -plane. Nodes are placed at $(0, 0)$, $(1, 0)$, $(0, 1)$, with edge nodes located at $(\alpha, 0)$, $(\gamma, 1 - \gamma)$ and $(0, \beta)$. The mapping to physical space is

$$\mathbf{x} = \sum_{i=1}^6 \mathbf{x}_i \phi_i(\xi, \eta), \quad (3.10)$$

where the cardinal shape functions are

$$\begin{aligned} \phi_2 &= \frac{1}{1 - \alpha} \left(\xi - \alpha + \frac{\alpha - \gamma}{1 - \gamma} \eta \right), \\ \phi_3 &= \frac{1}{1 - \beta} \eta \left(\eta - \beta + \frac{\beta + \gamma - 1}{\gamma} \xi \right), \\ \phi_4 &= \frac{1}{\alpha(1 - \alpha)} \xi(1 - \xi - \eta), \\ \phi_5 &= \frac{1}{\gamma(1 - \gamma)} \xi\eta, \\ \phi_6 &= \frac{1}{\beta(1 - \beta)} \eta(1 - \xi - \eta), \\ \phi_1 &= 1 - \phi_2 - \phi_3 - \phi_4 - \phi_5 - \phi_6, \end{aligned} \quad (3.11)$$

with parameters

$$\alpha = \frac{1}{1 + \left| \frac{\mathbf{x}_4 - \mathbf{x}_2}{\mathbf{x}_4 - \mathbf{x}_1} \right|}, \quad \beta = \frac{1}{1 + \left| \frac{\mathbf{x}_6 - \mathbf{x}_3}{\mathbf{x}_6 - \mathbf{x}_1} \right|}, \quad \gamma = \frac{1}{1 + \left| \frac{\mathbf{x}_5 - \mathbf{x}_2}{\mathbf{x}_5 - \mathbf{x}_3} \right|}. \quad (3.12)$$

The tangential vectors and metric coefficients are

$$\mathbf{e}_\xi = \frac{\partial \mathbf{x}}{\partial \xi} = \sum_{i=1}^6 \mathbf{x}_i \frac{\partial \phi_i}{\partial \xi}, \quad \mathbf{e}_\eta = \frac{\partial \mathbf{x}}{\partial \eta} = \sum_{i=1}^6 \mathbf{x}_i \frac{\partial \phi_i}{\partial \eta}, \quad (3.13)$$

$$h_\xi(\xi, \eta) \equiv \left| \frac{\partial \mathbf{x}}{\partial \xi} \right|, \quad h_\eta(\xi, \eta) \equiv \left| \frac{\partial \mathbf{x}}{\partial \eta} \right|, \quad (3.14)$$

and the unit normal and surface metric are

$$\mathbf{n}(\xi, \eta) = \frac{1}{h_S} \mathbf{e}_\xi \times \mathbf{e}_\eta, \quad (3.15)$$

$$h_S(\xi, \eta) = |\mathbf{e}_\xi \times \mathbf{e}_\eta|. \quad (3.16)$$

Unlike three-node flat triangles, neither the metric coefficients nor the unit normal remain constant over the six-node curved element. Surface integrals are evaluated as

$$\int_{\text{Triangle}} f(\mathbf{x}) dS = \int_0^1 \int_0^{1-\xi} f(\mathbf{x}(\xi, \eta)) h_S(\xi, \eta) d\eta d\xi. \quad (3.17)$$

3.3. Solid solver

The equations used for the solid solver employed in this work is described in detail in Section 2.4, to which the reader is referred for further understanding.

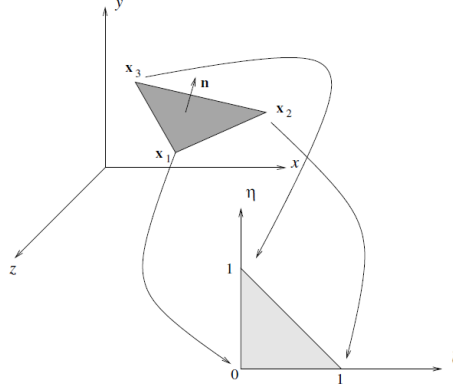


Figure 7: Triangle interpolation by (Pozrikidis 2002).

3.4. Validation

Validation is conducted against analytical and benchmark solutions to assess accuracy and robustness. Let $\mathcal{S}$ denote the mesh of quadratic (six-node) curved triangles discretising the boundary surface Γ . Define the mesh size h by

$$h := \max_{T \in \mathcal{S}} \text{diam}(T), \quad (3.18)$$

where $\text{diam}(T)$ is the element diameter. Assuming sufficient regularity and using a pointwise L^2 boundary norm, the BEM approximation u_h of the exact solution u satisfies

$$\|u - u_h\|_{L^2(\Gamma)} = \mathcal{O}(h^2), \quad \text{as } h \rightarrow 0, \quad (3.19)$$

consistent with quadratic elements.

3.4.1. Convergence of flow potential away from the body

We consider a single ellipsoid in an infinite, otherwise quiescent fluid, prescribed with initial linear and angular velocities. Analytical solutions for the resulting potential field follow from Lamb (1945). For the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad (3.20)$$

introduce ellipsoidal coordinates via

$$\frac{x^2}{a^2 - \theta} + \frac{y^2}{b^2 - \theta} + \frac{z^2}{c^2 - \theta} = 1, \quad (3.21)$$

and denote the three real roots by (λ, μ, ν) in increasing order. The coordinate λ measures the ellipsoidal radius. Following Lamb (1945), define

$$\alpha_0 = abc \int_0^\infty \frac{d\lambda}{(a^2 + \lambda)\Delta}, \quad \Delta = \{(a^2 + \lambda)(b^2 + \lambda)(c^2 + \lambda)\}^{1/2}. \quad (3.22)$$

For translation along x with speed U ,

$$\phi = Cx \int_\lambda^\infty \frac{d\lambda}{(a^2 + \lambda)\Delta}, \quad C = \frac{abc}{2 - \alpha_0} U. \quad (3.23)$$

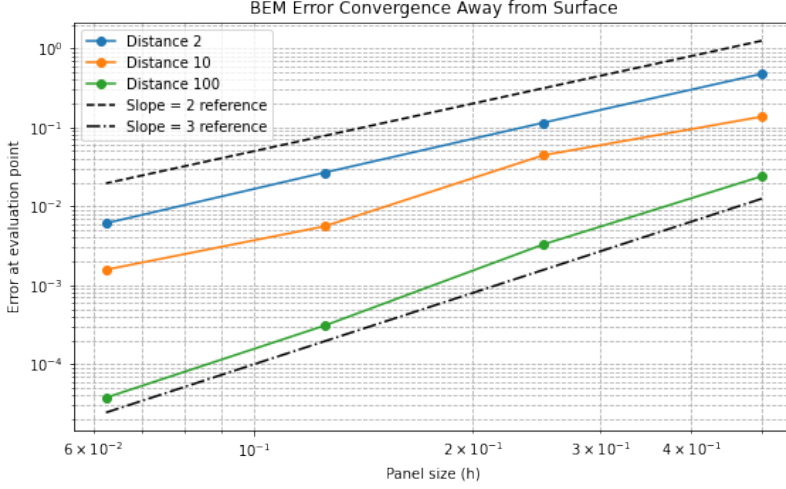


Figure 8: Convergence of the potential flow field of a single ellipsoid at selected points versus analytical solutions as panel size decreases. **[Are these considered to be acceptable levels of error?]**

356 For rotation about x with angular speed Ω_x ,

$$357 \quad \phi = Cyz \int_{\lambda}^{\infty} \frac{d\lambda}{(b^2 + \lambda)(c^2 + \lambda)\Delta}, \quad (3.24)$$

358 where

$$359 \quad C = \frac{(b^2 - c^2)^2}{2(b^2 - c^2) + (b^2 + c^2)(\beta_0 - \gamma_0)} abc \Omega_x, \quad (3.25)$$

360 and

$$361 \quad \beta_0 = abc \int_0^{\infty} \frac{d\lambda}{(b^2 + \lambda)\Delta}, \quad \gamma_0 = abc \int_0^{\infty} \frac{d\lambda}{(c^2 + \lambda)\Delta}. \quad (3.26)$$

362 These quadratures are inexpensive to evaluate numerically and, by linearity of Laplace's
363 equation, superpose for general rigid-body boundary conditions.

364 The results at distances 100, 10 and 2 from the ellipsoid centre, for initial $\|\mathbf{u}\| = \|\mathbf{\Omega}\| = 1$
365 in random directions and increasing mesh resolution, are shown in figure 8. As expected, the
366 convergence is $O(h^2)$ near the body and approaches $O(h^3)$ far from the body.

367 3.4.2. Periodic motion of a single ellipsoid

368 A single ellipsoid is placed in an inviscid fluid at rest at infinity and assigned initial linear
369 and angular velocities. The motion is integrated to $t = 10.0$ with $\Delta t = 10^{-4}$ over several
370 mesh resolutions, and compared with the original Kirchhoff equations (??); see figure 9. A
371 known periodic case is selected (convergence is not expected for chaotic trajectories). The
372 solver shows excellent agreement with the established motion.

373 3.5. Quantification of Chaos

374 We can use Recurrence Quantification Analysis (RQA) to quantify the trajectories and
375 orbits tracked by immersed bodies. The most fundamental part of RQA is a recurrence plot
376 (RP). An RP is a way to visualise the dynamics of a system, and when any given event
377 recurs. Importantly, it provides a method of visualising N-dimensional behaviour with two-
378 dimensional plots. This will only become more vital as we deal with higher and higher

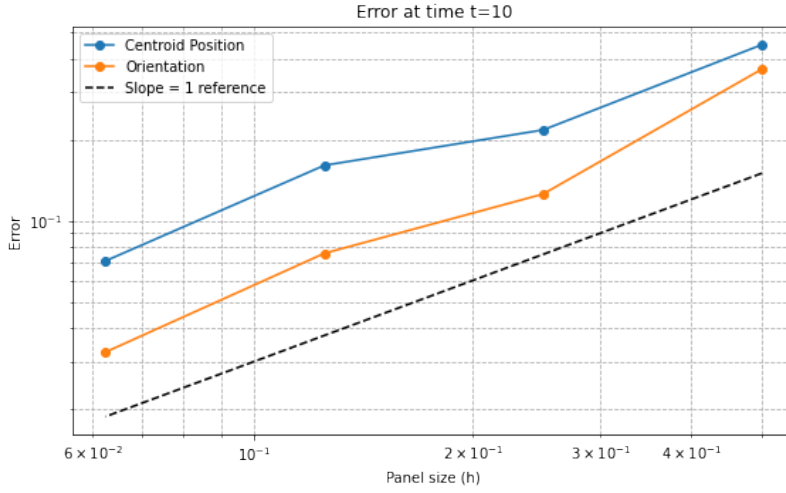


Figure 9: Convergence of periodic motion of an ellipsoid: extended Kirchhoff formulation versus original Kirchhoff equations.

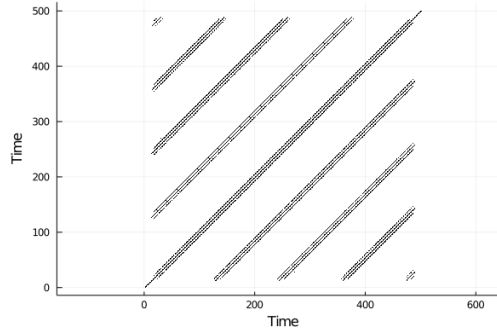


Figure 10: RQA plot of periodic behaviour at $\rho = 0.2, E = 4$.

379 dimensions as the number of bodies increases. The Recurrence matrix, R , is defined as:

$$380 \quad R_{i,j} = H(\eta - \|x(i) - x(j)\|) \quad (3.27)$$

381 where H is the Heaviside step function, R is an $N \times N$ matrix where N is the number of
 382 samples in the time series, and $x(i)$ is the position or orientation of the body at time i . η
 383 is the phase-space diameter, and represents how close two points must be to be considered
 384 recurred.

385 If a system is perfectly periodic, one would expect the RP to have solid diagonal lines
 386 at regular spaces (see Figure 10). If the system is quasiperiodic, there will still be diagonal
 387 lines, but these may be oddly spaced and segmented. Deterministic chaos is represented by
 388 "visually interesting" patterns in the plot (see Figure 11.) We can also define some useful
 389 metrics: Recurrence rate (RR) represents the probability of a solid's orientation or trajectory
 390 to be recurring:

$$391 \quad RR = \frac{1}{N^2} \sum_{i,j=1}^N R_{ij}. \quad (3.28)$$

392 We can also define DET as the determinism of the system, which is a measure of its

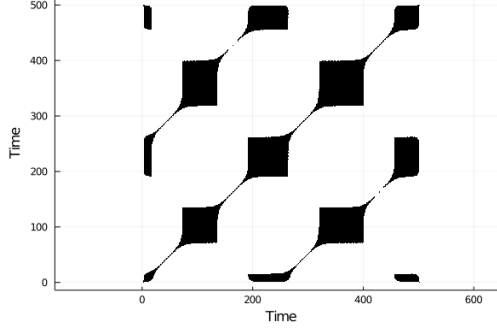


Figure 11: RQA plot of chaotic behaviour at $\rho = 4, E = 20$.

predictability, as follows:

$$DET = \frac{\sum_{l=l_{min}}^N lP(l)}{\sum_{l=1}^N lP(l)}, \quad (3.29)$$

where $P(l)$ is the frequency distribution of the lengths of the diagonal lines in the RP, l is the length of those lines, and l_{min} is the minimum acceptable length of points required to be considered a line. In this work l_{min} is set to 2. We also define

$$p(l) = \frac{P(l)}{\sum_{l=l_{min}}^N P(l)}, \quad (3.30)$$

and finally we can define the Shannon entropy ENTR:

$$ENTR = - \sum_{l=l_{min}}^N p(l) \ln(p(l)). \quad (3.31)$$

4. Multiple Solids

We investigate the dynamics of two immersed bodies in fluid flow as shown in figure 12, identifying conditions under which their interactions yield periodic, quasi-periodic, or chaotic behaviour. Focusing on the two-body problem isolates the fundamental mechanisms of fluid-mediated interaction without the additional complexity of many-body systems.

We vary the following three parameters:

- Density ratio, $\rho = \rho_s / \rho_f$, with ρ_s and ρ_f the solid and fluid densities.
- Energy ratio, $E = KE_{lin} / KE_{rot}$, the ratio of initial translational to rotational kinetic energy.

- Initial separation, d , with lengths normalised so the minimum body diameter is 1.

Both inviscid and viscous flows are considered. Inviscid cases impose non-penetration on each body and zero velocity at infinity. Viscous cases apply no-slip at body surfaces and periodic boundaries on a finite domain. While viscous runs are fewer due to the cost of resolving boundary layers and vorticity, they serve as valuable comparison points. The set-up is shown in Figure 12, with an example of the orbits of two bodies shown in Figure 13. In each simulation, each particle is given an initial linear and rotational velocity, such that their initial linear velocities are parallel, and their rotational velocities are randomised. This is illustrated in diagram The paths of the centroids of the bodies c_1, c_2 shall be referred to as trajectories, and the paths traced out by the orientations of the bodies according to $m_1 - c_1, m_2 - c_2$ shall be referred to as orbits.

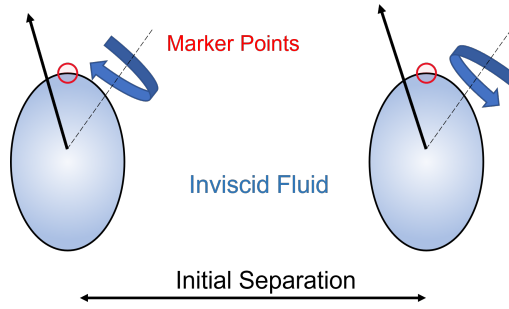


Figure 12: Set-up of the two-body problem. [Can you make the ellipsoids orient in different directions?]

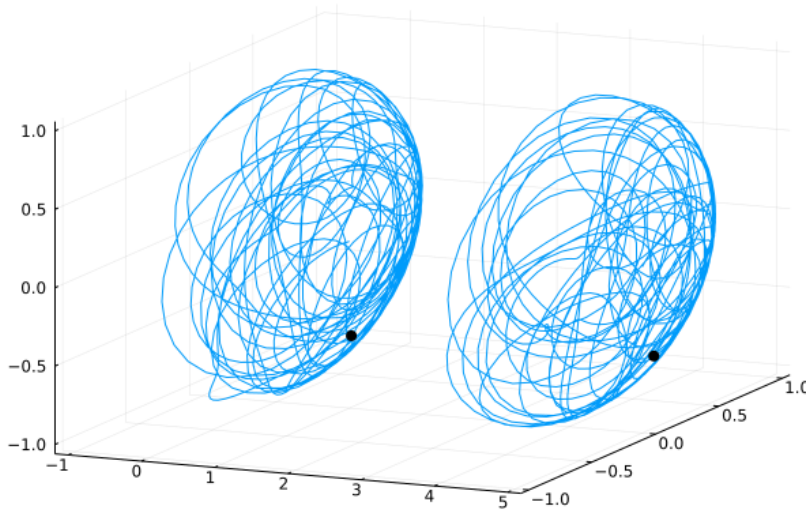


Figure 13: Orbits of two triaxial ellipsoids.

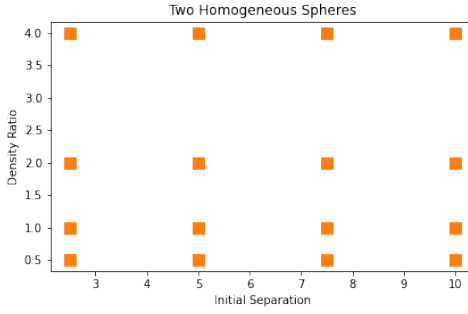
Geometry complexity is increased in stages: spheres, then spheroids, then triaxial ellipsoids. For each, we sweep key parameters to map regimes of non-trivial dynamics. The emergence of chaos without external forcing is a central theme; we show its sensitivity to geometry and initial energy distribution. Most simulations use POISE, with selected cases validated by (inviscid) X3D-NDOF as indicated.

4.1. Sphere

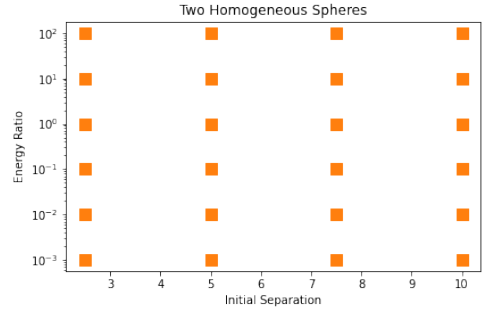
For two spheres in inviscid flow, orbits are necessarily periodic: the component of added-mass force perpendicular to the radial vector vanishes identically. Simulations corroborate this, as shown in Figures 14a and 14b, with representative periodic trajectories in Figure 15. Even strongly non-homogeneous initial energy partitions (Figures 16a–16c) do not break periodicity.

4.1.1. Non-homogeneous energy distributions

For each simulation of the immersed bodies, we can define k the ratio between the total kinetic energy of the first body, and the total kinetic energy of the second body. So far, only



(a) Two-sphere behaviour versus initial separation and density ratio. Orange squares: periodic. Simulations with POISE.



(b) Two-sphere behaviour versus initial separation and energy ratio. Orange squares: periodic. Simulations with POISE.

Figure 14: Two-sphere behaviour across parameter space: (left) density ratio variation and (right) energy ratio variation.

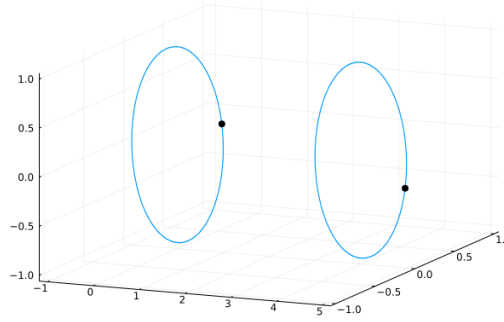
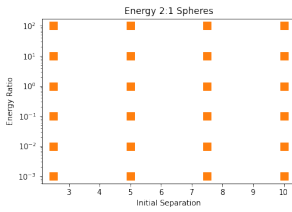
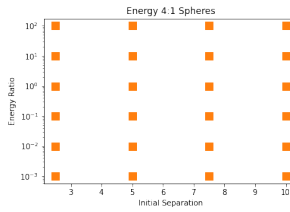


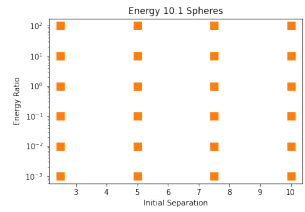
Figure 15: Example of periodic orbits for two spheres.



(a) E split 2:1.



(b) E split 4:1.



(c) E split 10:1.

Figure 16: Interaction behavior of two spheres for different energy splits. Orange squares denote periodic results from POISE.

435 the case $k = 1$ has been considered, but the following results have also been obtained with
 436 values of $k = 2, 4, 10$.

437

4.2. Spheroids

438 In this section, the behaviour of spheroids is analysed. For an ellipsoid with axes $a : b : c$,
 439 where $a \geq b \geq c$, it is a prolate ellipsoid if $b = c$, and an oblate ellipsoid if $a = b$. Both are
 440 considered below.

4.2.1. Interaction plots

We first delineate when the particles *interact* (i.e. each trajectory deviates perceptibly from that of the isolated-body counterpart). This is defined as follows. For a given two-particle simulation, we also compute the reference trajectories obtained when each particle is simulated alone with identical initial conditions (same shape, density ratio, energy ratio, initial orientation, and initial translational/rotational velocities). Let $\mathbf{X}_k(t)$ and $q_k(t)$ denote the centroid position and orientation quaternion of particle k in the two-particle case, and $\mathbf{X}_k^{(1)}(t)$ and $q_k^{(1)}(t)$ the corresponding single-particle reference solution. We classify the case as interacting if, over the observation window, either the maximum centroid deviation exceeds a prescribed tolerance,

$$\max_{t \in [0, T]} \|\mathbf{X}_k(t) - \mathbf{X}_k^{(1)}(t)\| > \varepsilon_X,$$

or the maximum orientation deviation exceeds a prescribed tolerance,

$$\max_{t \in [0, T]} d(q_k(t), q_k^{(1)}(t)) > \varepsilon_q,$$

for at least one particle $k \in \{1, 2\}$. Here $d(\cdot, \cdot)$ is an orientation-distance metric, defined as follows:

To quantify the difference between two orientations, we define an orientation-distance metric on the rotation group $SO(3)$. If the particle orientation is represented by a unit quaternion $q = (q_0, \mathbf{q}_v)$, the relative rotation between two orientations q_1 and q_2 is given by

$$q_{\text{rel}} = q_1^{-1} q_2,$$

where q_1^{-1} denotes the quaternion conjugate (for unit quaternions). The magnitude of the misalignment is then defined as the principal rotation angle associated with q_{rel} ,

$$d(q_1, q_2) = 2 \arccos(|\langle q_1, q_2 \rangle|),$$

where $\langle q_1, q_2 \rangle$ denotes the Euclidean inner product of the quaternion components. The absolute value ensures invariance under the double-cover property (q and $-q$ represent the same physical orientation). This metric yields the smallest rotation angle required to align the two orientations and lies in the range $[0, \pi]$, and $\varepsilon_X, \varepsilon_q$ are chosen to exceed numerical error determined from single-body convergence tests. For the purposes of this work, ε_X was selected to be 0.01, and ε_q was selected to be 0.01. Figure 17 shows that interactions arise at large separations when $E \gtrsim 10$, and at small separations even for $E \sim 10^{-3}$. Relative to spheres, spheroids interact more readily at close separations, reflecting enhanced rotational energy transfer to the fluid.

4.2.2. Parameter analysis

Chaotic dynamics for *single* spheroids do not occur, as the equations are integrable unless triaxial (Kozlov & Onishchenko 1982). The two-body interaction introduces sufficient complexity for chaos to emerge. Figures 19–20 summarise results; an example chaotic orbit appears in Figure 18.

4.2.3. Departures from sphericity

Defining ϵ as a small change in the axes, let prolate axes be $(1 + \epsilon)^2 : (1 + \epsilon)^{-1} : (1 + \epsilon)^{-1}$ and oblate axes $(1 + \epsilon) : (1 + \epsilon) : (1 + \epsilon)^{-2}$, preserving volume as $\epsilon \in [0, 1]$. Figure 22 shows determinism versus ϵ : increasing asphericity promotes chaos, with a brief transitional window near $\epsilon \approx 10^{-3}$. Both homogeneous (both particles vary) and heterogeneous (one fixed at $\epsilon = 0$ or 1) cases follow the same trend, with the partner's shape shifting the onset by

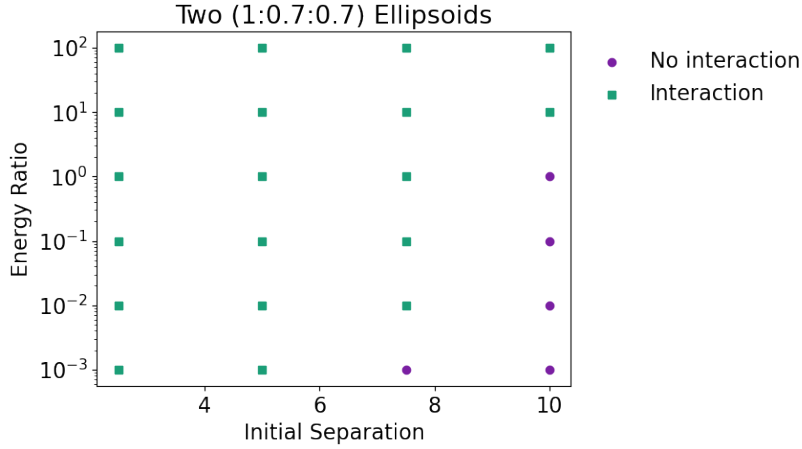


Figure 17: Interaction map for two spheroids as initial separation and energy ratio vary. Density ratio $\rho = 2$.

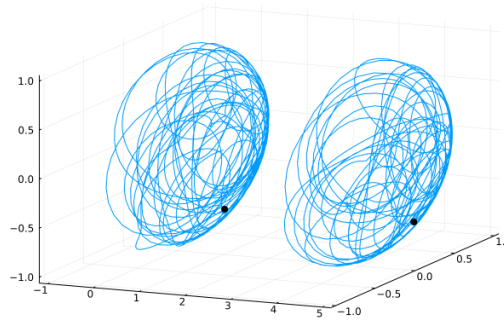


Figure 18: Representative chaotic orbits for two spheroidal bodies.

roughly one order of magnitude. Quasi-periodicity for a nearly spherical partner is illustrated in Figure 21.

4.3. Triaxial ellipsoids

A triaxial ellipsoid has distinct principal axes $a \neq b \neq c$.

4.3.1. Interaction plots

Interaction trends broadly mirror the spheroidal case: larger E and smaller d favour interaction (Figure 23). Triaxiality increases the propensity to interact at lower E than spheres or spheroids, although the difference from spheroids is modest in our sampling. The effect is consistent with higher linear speeds at fixed E reducing effective separation over a turnover time.

4.3.2. Parameter analysis

A parameter sweep over d , E , and ρ (Figures 24–25) shows a substantially broader chaotic regime than for spheroids. Loss of symmetry in triaxial shapes permits chaotic dynamics even in the absence of multi-body interaction, leading to chaotic behaviours even in situations where separation is large.

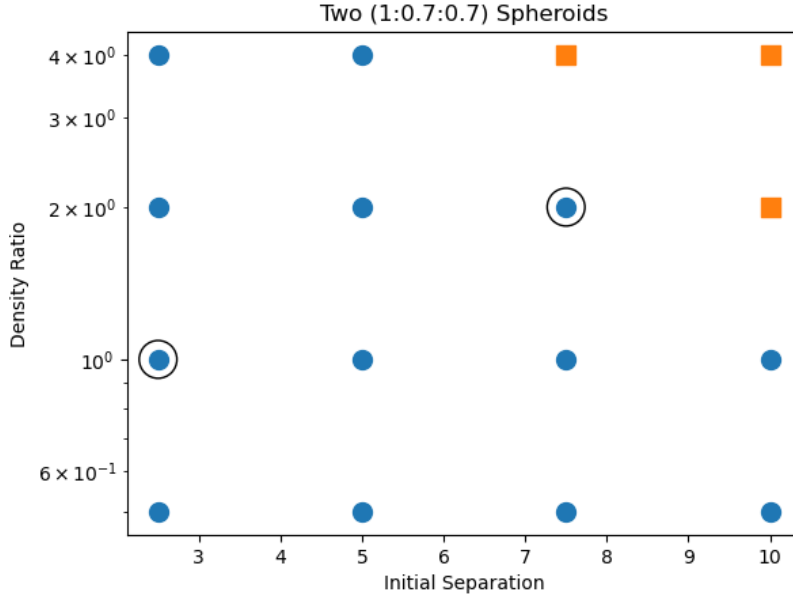


Figure 19: Two-spheroid behaviour versus initial separation and density ratio ($E = 1$).
Orange squares: periodic; blue circles: chaotic; black circles: cases validated with X3D-NDOF POISE.

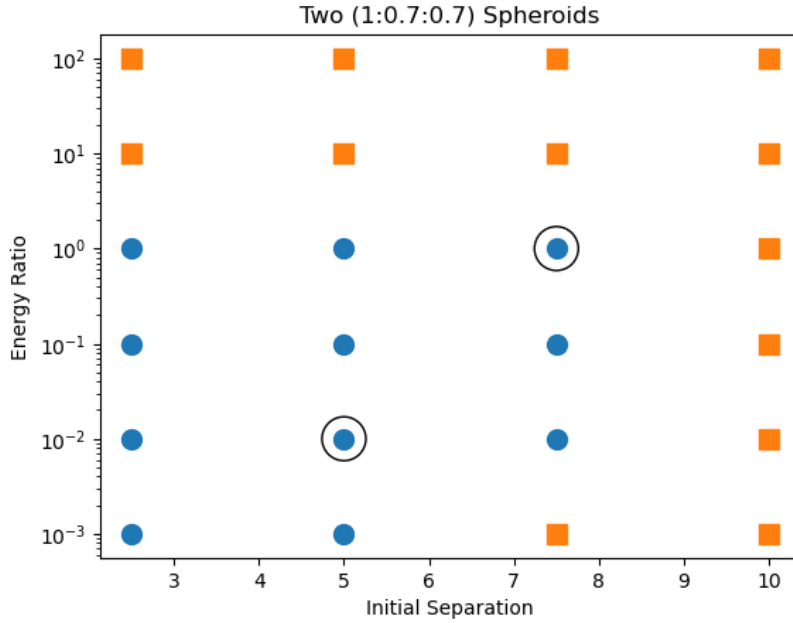


Figure 20: Two-spheroid behaviour versus initial separation and energy ratio ($\rho = 1$).
Orange squares: periodic; blue circles: chaotic; black circles: cases validated with X3D-NDOF POISE.

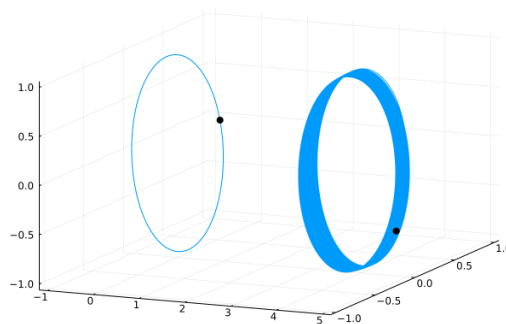


Figure 21: Quasi-periodic motion of body 2 with spherical partner; $\epsilon = 10^{-3}$.

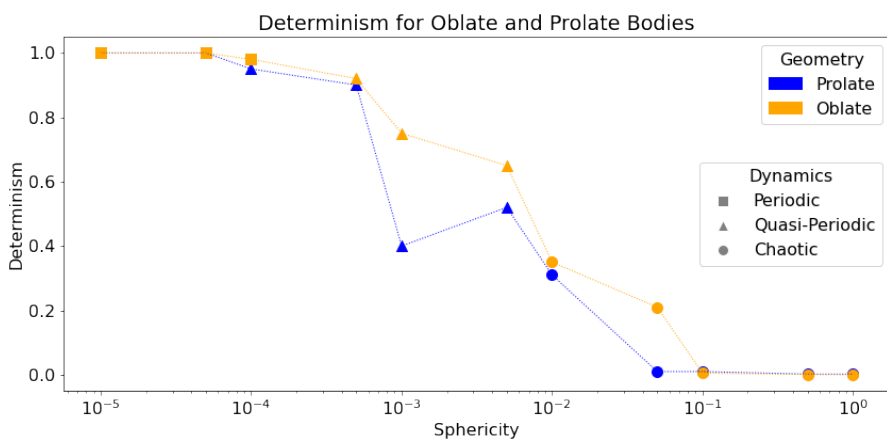


Figure 22: Determinism versus departure from sphericity for two spheroids. Dynamics are represented by marker shape. POISE.

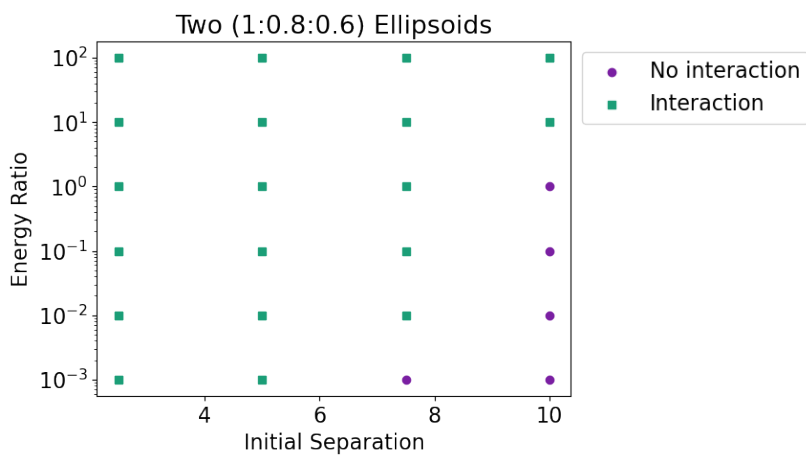


Figure 23: Interaction map for two triaxial ellipsoids.

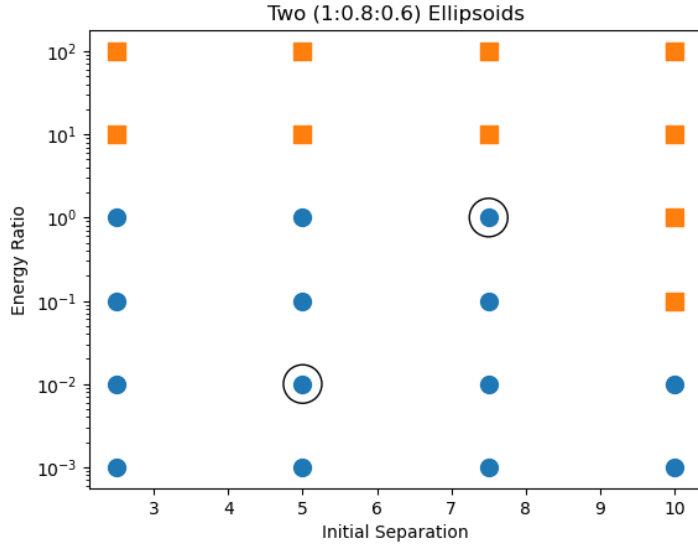


Figure 24: Two-ellipsoid behaviour versus initial separation and energy ratio. Orange squares: periodic; blue circles: chaotic; black circles: X3D-NDOF validations. POISE.

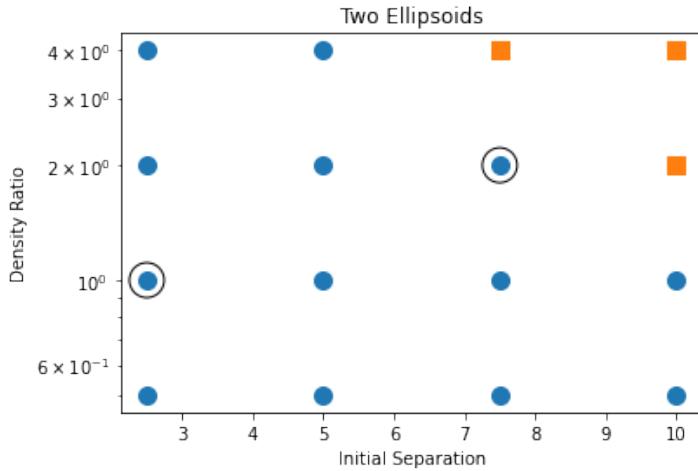


Figure 25: Two-ellipsoid behaviour versus initial separation and density ratio. Orange squares: periodic; blue circles: chaotic; black circles: X3D-NDOF validations. POISE.

4.3.3. Departing from sphericity

Let axes be $(1 + \epsilon) : 1 : (1 + \epsilon)^{-1}$ so that $\epsilon \in [0, 1]$ spans sphere to highly triaxial. Figure 27 shows periodicity at $\epsilon \lesssim 10^{-4}$ and robust chaos for $\epsilon \gtrsim 10^{-2}$, with earlier transition when the companion is maximally aspherical ($\epsilon = 1$). Figure 26 illustrates a triaxial body driven chaotic by a spherical neighbour.

4.4. Viscous effects

Essmann et. al. showed that at high Reynold's number, the viscous effects do not dominate the behaviour, and the presence of chaos is still determined by added mass effects which are seen in the inviscid scenario. Using X3D-NDOF, we simulate two-body viscous flows at $Re_p \sim$

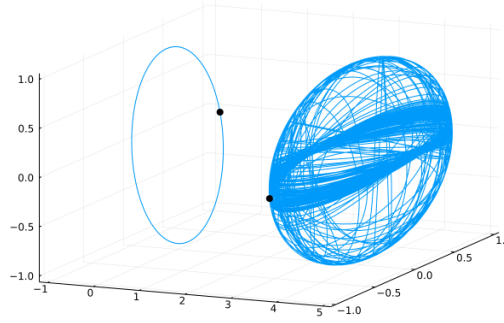


Figure 26: Chaotic trajectory of a triaxial ellipsoid with $\epsilon = 10^{-2}$ interacting with a spherical particle.

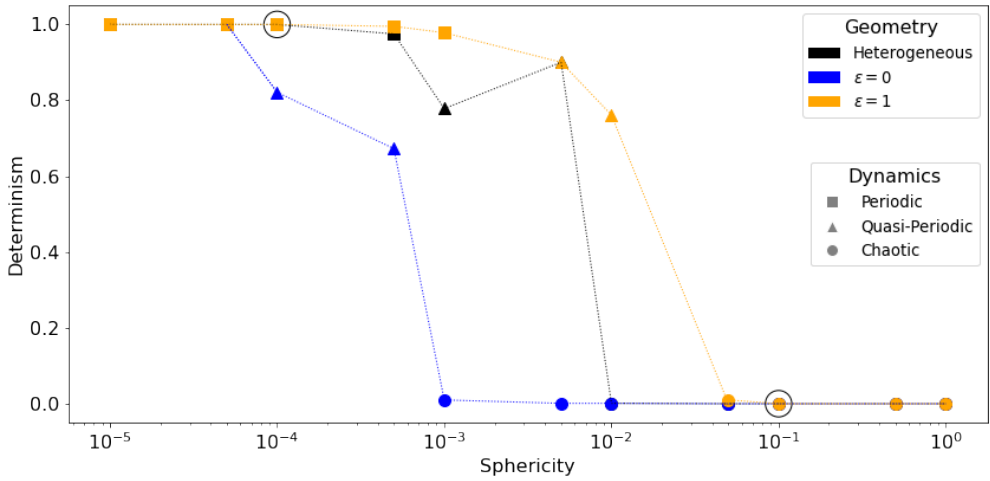


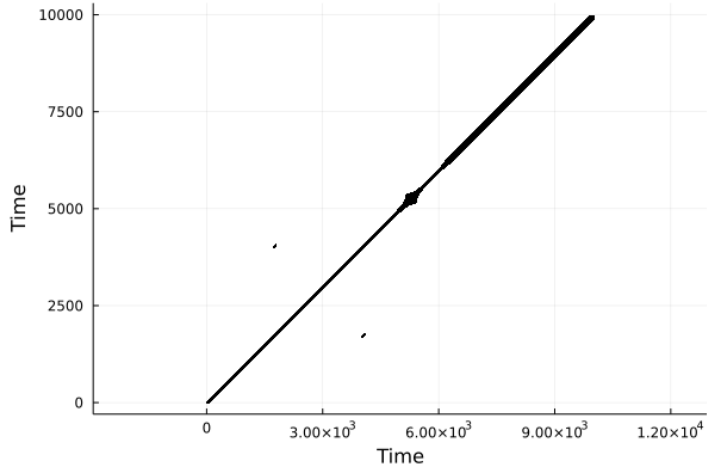
Figure 27: Determinism during homogeneous/heterogeneous departures from sphericity. POISE.

100–300 in periodic domains. Compared with prior single-body studies at $Re \sim 10^4$ – 10^5 , our resolution is limited by the fixed grid near bodies (no adaptive mesh), restricting attainable Reynolds numbers. Early-time behaviour remains dominated by added-mass effects, as in inviscid flow. At later times, viscous damping grows and vortex shedding could trigger chaos, but conclusive onset is unresolved within practical domain sizes and time horizons. Recurrence plots in Figure 28 compare inviscid single-body, viscous single-body, and two-body viscous cases, evidencing rising dynamical complexity with viscosity and interactions.

514

4.5. Conclusion

Two-body inviscid sphere systems are constrained to periodic motion by symmetry, even under extreme heterogeneity of energy and density ratios. Relaxing symmetry—via spheroidal or triaxial shapes—substantially broadens the regimes exhibiting chaos once fluid-mediated interaction is present, with triaxiality most susceptible. Viscosity further enriches dynamics and, together with multi-body effects, promotes irregularity, suggesting the potential for vortex-induced chaos given larger domains and longer integrations. These results clarify how geometry, energy distribution, and viscosity govern the onset of complexity in coupled



□

(a) Recurrence plot: single body, inviscid.

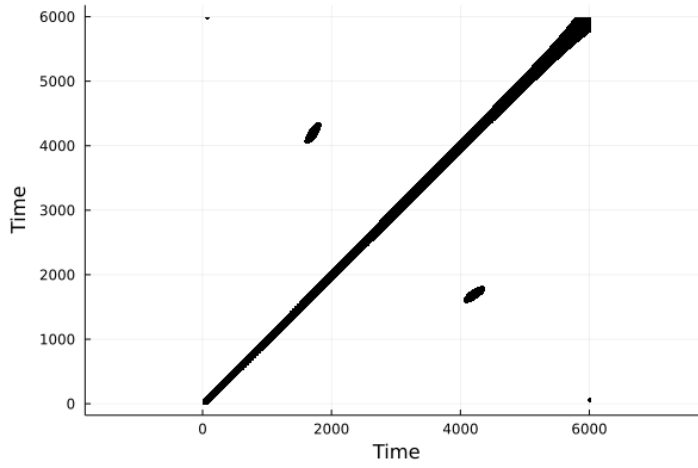
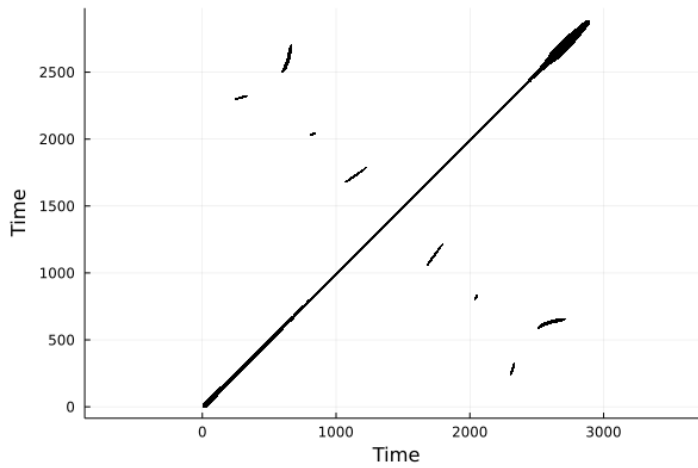
(b) Recurrence plot: single body, viscous ($Re_p = 100$).(c) Recurrence plot: two bodies, viscous ($Re_p = 100$).

Figure 28: Recurrence plots across flow regimes and configurations.

fluid–solid systems, and they provide a basis for scaling to many-body and higher–Reynolds-number regimes.

5. Conclusion

[We need to show typical trajectories of two spheres and show how they are periodic. Also dont we have any chaotic cases where both trajectories occupy the same region in space, or have some overlap? Also we have said that our method is validated for viscous cases, but we do not show any effect of viscosity. Do you want to do that separately? I would vote for at least one case where viscosity changes things and one where it does not.] This study has investigated the dynamics of two freely moving rigid bodies in fluid flow, focusing on the conditions under which motion transitions from regular to chaotic. By combining a boundary element method for inviscid potential flow (POISE) with a direct numerical simulation framework employing an immersed boundary method (X3D–NDOF), we systematically explored the effects of density ratio, energy ratio, initial separation, and body geometry.

The results show that spheres in inviscid flow remain strictly periodic, even under large contrasts in energy or density, reflecting the integrable nature of their dynamics. In contrast, spheroidal particles—although individually integrable—exhibit chaotic trajectories when coupled in pairs, as fluid-mediated interactions introduce sufficient complexity to destabilise the motion. Triaxial ellipsoids display the widest and most robust chaotic regimes, underscoring the role of geometric asymmetry in promoting irregular behaviour. Departures from sphericity accelerate the onset of chaos, with heterogeneous pairs such as a sphere and an ellipsoid showing irregular motion at lower thresholds than homogeneous configurations. Complementary viscous simulations at moderate Reynolds numbers further reveal that viscosity damps early-time dynamics but enhances long-term irregularity through vortex shedding.

Together, these findings demonstrate that even the minimal two-body problem contains unexpectedly rich dynamics once symmetry is broken. The emergence of chaos in inviscid regimes challenges the perception of potential flow as inherently simple and highlights the importance of nonlinear effects in fluid–structure interactions. Future work will extend this framework to larger particle ensembles, higher Reynolds numbers, and adaptive numerical methods, linking the insights from two-body dynamics to the collective behaviour of suspensions in natural and engineered flows.

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